MIL-C-684D(MU)
29 January 1971
SUPERSEDING
MIL-C-684C(MU)
23 December 1966

MILITARY SPECIFICATION

CUPS, BULLET, JACKET, COPPER ALLOY CLAD STEEL; FOR SMALL ARMS AMMINITION

1. SCOPE

1.1 This specification covers annealed copper alloy clad steel bullet jacket CUPS used in the manufacture of bullets for small arms cartridges.

2. Applicable Documents

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATION

Military

MIL-C-3993 - Copper and Copper-Base Alloy Mill Products; Packaging of

All cartridge specifications specified in the contract or purchase order and Appendix "A", (see 6.1)

STANDARDS

Federal

Federal Test Method Standard No. 151- Metals; Test Methods

FSC 1305

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2.1 (Cont'd)

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipnent and Storage
MIL-STD-1232	- Visual Inspection Standards for Cups and Disks used in Small Arms Ammunition Manufacture

DRAWINGS

All bullet jacket cup, bullet and cartridge drawings specified in the contract or purchase order and Appendix "A", (see 6.1)

U.S. Army Munitions Command

D7553671	- BOX, Wirebound, Shipping for Small Arms Ammunition Cups and Disks for Bullet Jackets and Cases - List of Parts and Assembly
F7553844	- Box, Shipping for Small Arms Ammunition Cups and Disks for Bullet Jackets and Cases - List of Parts and Assembly

PUBLICATION

AMSMU-P 715-501FA1	- Ammunition Ballistic Acceptance Test Methods, Test Procedures for 5 .56MM Cartridges

(Copies of specifications, standards , drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contacting officer.)

3. REQUIREMENTS

3.1 General. - The bullet jacket cup shall comply with the requirements specified on the applicable drawing, the referenced specifications and the following. Cups intended for qualification of product and source of supply shall comply with the requirements cited below in paragraphs 3.3, Hardness; 3.6, (Coating; and 3.7, workmanship (see 4.2). Cups Intended for manufacture of small arms bullet jackets shall be products which have been tested and passed the qualification tests specified herein and have been listed on, or approved for listing on the applicable qualified products list (QPL) (see 6.3); these cups shall comply with all following requirements. In addition, the cups intended for manufacture of small arms bullet jackets shall be representative of the materials, process, production methods, facilities and workmanship reflected in the product qualification material.

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3.2 Qualification. - The cups furnished under this Specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.3 Hardness.- The hardness of the steel cup, after removal of the copper alloy cladding, shall not exceed Rockwell B50, 49.5 (R30T) or 77 (R15T). The range of individual hardness determinations shall not exceed fifteen (15) integers of measurement on the Rockwell B scale.

3.4 Workability .- The bullet jacket cup shall be suitable for processing into a bullet jacket complying with the applicable bullet drawing and referended specifications. The method of processing shall be similar to the established method, the number of individual processes (drawing, annealing and forming operations) shall not exceed the established number of operations (see 6.4). Note: The appropriate cup, bullet and cartridge drawings are cross-referenced in Appendix "A".

3.5 Performance .- When assembled into the designated cartridge and test fired at the instrumental velocity cited in the applicable cartridge specification, the processed bullet jacket cup shall maintain the bullet configuration throughout its in-flight trajectory (see Appendix “A”).

3.6 Coating.-The cup shall be coated with a residual film of neutral soap.

3.7 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Cladding.-The thicker layer of copper alloy cladding shall be on the outer surface of the cup.

3.7.2 Metal defects. - The cup shall be free of laminations, slivers, laps, crack, scale, scratches, dents, bent sidewalls, V’d edges and wire edges. The cup shall have continuous layers of copper alloy cladding over the outer and inner surfaces of the cup except at extreme edge at mouth of cup .

3.7.3 Foreign matter.-The cup shall be clean and free of oil, grease , dirt, grit, oxidation, stains, chips and dichromate residue.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unlesss otherwise specified in the contract or purchase order,the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the Inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions .-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of inspections .- The examinations and tests required to assure conformance of these items or lots of items to be offered for acceptance are classified as follows:

a. Qualification inspection
b. First article inspection
c. Lot acceptance inspection

4.3 Qualification .- Qualification inspection shall consist of the examinations and tests specified for all requirements of this specification. The workability y and performance tests shall be performed by a Government approved facility.

4.3.1 Qualification samples .-A minimum of 1,010,000 bullet jacket cups shall be submitted qualification approval. The sample cups shall be finished products, representative of the materials and workmanship to be applied in normal production by the manufacturer and shall comply with requirements of 3.1. These cups shall be tested in two phases, the initial 10,000 cups shall be tested as Phase I of the qualified products list qualification; these cups shall meet all the requirements and tests to qualify the 1,000,000 cups for the Phase II or final qualified products list qualification tests.

Qualification Sample Sizes

Table I

		Quantity of Cups Required	
Cup Dwg. No.	Type	Phase I	Phase II
B7553579	7.62MM Ball M80	10,000	1,000,000
B7553579	Cal. .30 Ball M2	(1)	1,000,000
B10542383	Cal. .30 Carbine Ball	(1)	1,000,000
B7553946	7.62MM Tracer M62	10,000	1,000,000
B7553946	Cal. .30 Tracer M25	(2)	(2)
B10542387	Cal. .50 Tracer M17	(3)	10,000 (3)
B11731734	5.56MM Tracer M196	10,000	1,000,000
BX10542421	5.56MM Ball M193	10,000	1,000,000

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4.3.1 (Cont'd)

(1) Phase I qualification is granted on the basis of supplier's compliance with Phase I 7.62MM Ball M80 cup requirements; otherwise 10,000 cups are required.

(2) Compliance with Phase I and II 7.62MM Tracer M62 cup requirements will qualify supplier; otherwise 10,000 Phase I and 1,000,000 Phase II cups are required.

(3) Qualification limited to total compliance of one 10,000 cup sample with all requirements.

4.3.2 Qualification approval.-Facilities submitting bullet jacket cups for Qualified Product List (QPL) shall be placed on the QPL for the applicable bullet jacket cup design if the samples submitted meet the requirements of Section 3 of this specification and the conditions of Table I. Failure to meet all the requirements shall be cause for refusal to grant qualification approval.

4.4 First article.

4.4.1 Initial production sample. -At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5200 cups. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production.

4.4.2 Examination and test. -After inspection and provisional acceptance cups of the sample at source, 200 cup shall be inspected for all requirements of the drawing and specifications at a Government laboratory or such other facility specified in the contract. The remaining 5000 cups shall be subjected to a work test, and in turn 300 samples thereof shall be assembled into cartridges and subjected to the performance test. (see 6.1.7)

4.4.3 Initial production sample failure .-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.5 Lot acceptance inspection.

4.5.1 Submission of product. -The product shall be submitted in accordance with the "Submission of Products" provisions of MIL-STD-105.

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4.5.2 Lot identification. -Each lot of cups shall be identified as to type, caliber, and model as well as with a lot number and the supplier's identification as assigned by the procuring activity.

4.5.3 Examination. -Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table II, using applicable sampling plan and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent. and the AQL for the minor class shall be 2.50 percent. All nonconforming components shall be rejected.

4.5.3.1 Classification of defects. - The classification of defects shall as specified in Table II.

Table II

Defect		Method of Inspection
MAJOR		
101.	Defective cladding	visual
102.	Improperly formed or mutilated	Visual
103	Laminations or laps	visual
104.	Scratches	visual
105	Cladding thickness 1/	Gage
106.	Outside diameter	Gage
107.	Base thickness	Gage
108.	Wall thickness	Gage
109.	Wall thickness variation	Gage
110.	Wall height variation	Gage
111.	Weight	scale
MINOR		
201.	Oil, grease, dirt	Visual
202.	Dents or bent edges	visual
203.	Oxidation	Visual
SPECIAL		
301.	Absence of neutral soap film	Certification
302.	Dichromate residue	Certification

1/ See 3.7.1

NOTE: Refer to applicable gliding metal clad steel cup section of MIL-STD-1232 for visual defect standards for visual defects listed in Table II.

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4.5.4 Tests .-The following tests shall be conducted in accordance with method and procedures specified in 4.6.

4.5.4.1 Hardness. -A sample of 25 cups shall be tested for hardness. Failure of any cup to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.5.4.2 Workability. -The qualification samples specified in Table I, the sample of 5000 cups from the initial production sample, and a sample of 5000 cups from each production lot shall be subjected to a workability test (see 6.1.7). Occurrence of any punch-outs, point diameter variations or other failure to comply with 3.4 shall be cause for rejection of the sample or lot as applicable.

4.5.4.3 Performance. -A sample of 300 bullet jackets, fabricated under 4.5.4.2, shall be assembled into cartridges; these shall be tested for stripping. Occurrence of stripping on any bullet shall be cause for rejection of the lot.

4.5.5 Packaging, packing and marking inspection. -Inspection for packaging, packing and marking shall be as directed by the contracting officer.

4.5.6 Inspection equipnent. - The examinations shall be made using commercial measuring equipment with a precision of 10 percent of the tolerance on the product dimension.

4.6 Test methods and procedures.

4.6.1 Hardness. -Each cup selected for this test shall be sectioned longitudinally The copper alloy cladding shall be removed and any surface effect removed by grinding or polishing. The specimen shall then be mounted on a flat anvil with the concave surface upward and readings taken near the center of the sidewall. The test shall be conducted In accordance with Method 243.1 in Federal Test Methods Standard No. 151

4.6.2 Workability. -The workability test shall comprise all of the process production operations used in the manufacture of the bullet jacket (ace 6.4).

4.6.3 performance.-The stripping test shall be performed in accordance with AMSMU-P 715-501FA1.

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5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A (Worldwide shipment). - The cups shall be packed In accordance with MIL-C-3993. Gross weight of each pallet type box with cups shall not exceed 2000 pounds.

(5. 1.2 Level C Domestic shipment). -The cups shall be packed in pallet type boxes conforming to Drawings D7553671 and F7553844. Gross weight of each pallet type box with cups shall not exceed 3000 pounds. Only one type of box shall be used for any one lot of cups.

5.2 Marking.- Each box shall be marked with the name of the material, type of cup, lot number, name of contractor, and the contract or order number in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. -Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Drawing number of cup and designated bullet type.

6.1.3 Classification of product .-For “qualification” or manufacture of Small Caliber Ammnition.

6.1.4 Level of packing.

6.1.5 Provision for submission of acceptance Inspection reports, in duplicate, to the contracting officer, containing final Inspection remits for each lot of cups presented to the Government

6.1.6 Provision for inspection of Packaging, Packing and Marking.

6.1.7 Provision for conduct of workability and performance test. -Where feasible, the workability and performance test portion of the initial production and lot acceptance sample should be forwarded to the facility designated to receive the procured cups. That facility should perform the workability test using tools, drawing and cleaning compounds, heat treating processes, lubricants and equipment In accordance with that facility's manufacturing process. The remainder of the initial production sample should be forwarded to a Government approved facility for inspection, The workability and performance tests should not be perfomed until acceptability of the cups has been determined in the inspection phase and instruction issued by the Government to proceed with the test. In those instances where cups are being procured for delivery to multiple consignees, these tests shall be performed in a Government approved facility as specified in the contract or purchase order.

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6.2 The supplier shall certify that each qualification or production lot of cups is free of all traces of bichromate, and is coated with a residual film of neutral soap.

6.3 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is U.S. Army, Frankford Arsenal, Philadelphia, Pa. 19137 and information pertaining to qualification of products may be obtained from that activity.

6.4 Data pertaining to methods of processing the bullet jacket cups into finished bullets are available for examination at Frankford Arsental or in the office of the Procuring Agency.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project No. 1305-A666

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APPENDIX "A"

Bullet Type	Bullet Jacket Cup	Drawing Number Bullet	Cartridge	Cartridge Specification
5.56MM Ball, M193	BX10542421	CX10542369	D10523632*	MIL-C-9963
5.56MM Tracer, M196	B11731734	C10542726	C10534193*	MIL-C-60111(MU)
7.62MM Ball, M80	B7553579	B8595669	C10521998	MIL-C-46931(MU)
7.62MM Tracer, M62	B7553946	C7553758	C10522000	MIL-C-46281(MU)
Cal. .30 Ball, M2	B7553579	B6171958	C6137544	MIL-C-1313(MU)
Cal. .30 Carbine Ball, M1	B10542383	C6200955	C6200954	MIL-C-3369(MU)
Cal. .30 Tracer, M25	B7553946	C7640668	C7640667	MIL-C-1317
Cal. .50 Tracer, M17	B10542387	C7638551	D7672165	MIL-C-1318

* Copper Alloy Clad Steel Jacketed bullet is to be substituted in place of the copper alloy jacketed bullet.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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